

Stereotypes and Memory*

Huen Tat Au-Yeung

Zhuokun Liu

November 18, 2025

Abstract

Imperfect memory can sustain widely held stereotypes that are inaccurate. To establish causal evidence, we conduct an online experiment where subjects grade a gender-stereotyped quiz for a class of students and later predict their performances on a similar quiz. We vary (1) reliance on memory by manipulating access to grading records and (2) the class with different gender performance gap. We find that stereotypes—perceived gender performance gaps—persist more when subjects rely on memory, but only when the true performance contradicts stereotypes. This result holds only for group-level, but not for individual-level assessments. Then, we extend the standard Bayesian framework with imperfect memory and find evidence of *memory attenuation*: reliance on memory substantially reduces how accurately individuals incorporate previously observed performance into their beliefs.

*Huen Tat Au-Yeung: IIES, Stockholm University, huen-tat.au-yeung@iies.su.se; Zhuokun Liu: Goethe University Frankfurt, zhuokun.liu@econ.uni-frankfurt.de. We are grateful for valuable comments from Johannes Abeler, Anna Dreber Almenberg, Peter Andre, Zoë B. Cullen, Jonathan de Quidt, Mitchell Downey, Nicola Gennaioli, Michael Kosfeld, Heather Sarsons, Frederik Schwerter, Ferdinand von Siemens, and participants at the Verona Experimental Meeting 2024, the “Choice and Welfare: Beyond the Rational Agent” Workshop, Young Economists’ Meeting 2025, the ESA World Meeting 2025, the BEAT Conference 2025, the SDM Conference 2025, the ESA European Meeting 2025, and seminar participants at the IIES and SOFI Brown Bag Series, Central University of Finance and Economics, and Jinan University. Financial support from the Leibniz Institute for Financial Research SAFE and Goethe University Frankfurt is gratefully acknowledged. IRB approval was obtained from the Joint Ethics Committee of the Goethe University Frankfurt and the Johannes Gutenberg University Mainz. The experiment was pre-registered on AsPredicted [#205685](#).

1 Introduction

Discrimination is documented in different settings and is highly persistent. Economists have long been interested in understanding its source. In the earlier days, discrimination is seen as some personal tastes (Becker, 1957), or actual statistical differences among groups (Aigner and Cain, 1977; Phelps, 1972). More recently, economists have focused on inaccurate beliefs as a source of discrimination (Bohren et al., 2023), which can result from stereotypes (Bordalo et al., 2016). Understanding why such inaccurate beliefs persist is crucial for welfare analyses and for designing policies aimed at correcting these beliefs and reducing discrimination.

One explanation for the persistence of inaccurate beliefs is imperfect memory (Bordalo et al., 2021, 2022). Drawing on insights from psychology, the literature models belief formation based on recalled experiences, highlighting that some are easier to recall than others. This selective recall can distort beliefs, as not all information from past experiences is used.

This paper studies how imperfect memory sustains stereotypes—widely held beliefs about groups—that are inaccurate. We focus on two key processes in belief formation: encoding and retrieval. Individuals first encode experiences into their memory, typically involving members of different groups, over time. Later, when forming beliefs about a group, they retrieve relevant past experiences with members from the group. This suggests that stereotypes can persist due to either selective encoding or selective retrieval of information.

We conduct an online experiment to distinguish between selective encoding and selective retrieval in sustaining stereotypes. We find that stereotypes persist more when individuals rely on memory rather than being reminded of the information, but only when the stereotypes are inaccurate. This persistence is driven by selective retrieval, as individuals fail to recall information accurately. Notably, this effect occurs for group-level beliefs but not for beliefs about individuals within the group, suggesting that the encoding of individual information is not selective.

To further explain our findings, we extend the Bayesian framework to incorporate imperfect recall. Instead of relying on accurate information, we model individuals using recalled information, which may be both contaminated and noisy. These memory imperfections hinder effective belief updating, contributing to the persistence of stereotypes.

In the experiment, our subjects perform a grading task for a gender-stereotyped quiz about cars. This topic is generally perceived as one where males perform better than females. We first elicit the prior beliefs, and then grading starts. We provide the answers for the quiz to our subjects to ensure correct memory encoding and minimize manipulation in the evaluation. The grading task creates a memory database on the performance of the quiz-takers (*students*) in a natural manner.

We use a 2×2 between-subject design to vary (1) the gender performance gap in the graded quiz and (2) reliance on memory. Subjects are randomly assigned to grade one of two classes, each consisting of 10 males and 10 females. The classes differ in their gender performance gap: in the *Female-advantaged class*, females outperform males on

average, whereas in the *Male-advantaged class* the opposite direction holds. We also manipulate reliance on memory by controlling access to grading records (*Hide/Show*). For subjects in the Hide condition, graded scores are hidden, requiring them to rely on memory for subsequent decisions. The remaining subjects can see the graded scores when making decisions.

After grading, our subjects first make several predictions for a similar quiz, which we interpret as the posterior beliefs about the students' abilities; then they are asked about the student performance in the quiz they just graded, which we refer to as recalls. We focus on beliefs and recalls of the exact same class of students they graded. As our subjects do not know these students, we control for the experiences with the students directly.

For both quizzes, we ask for the average performance by gender, and the performance of two individuals from different gender groups. We pick the students who score either 0 or 5, i.e., the best or the worst performers (*outliers*). After that, our subjects nominate a student they believe performs the best in the quiz.

As we focus on the domain where males are generally perceived to perform better than females, we define the stereotype in our experiment as the subjective gender performance gap (gender gap) as follows:

$$\text{Gender gap} := \text{average male performance} - \text{average female performance}$$

Our design has three unique features.

First, we examine memory in a real-life context by controlling subjects' direct experiences with the students in the experiment, while allowing for the potential influence of prior experiences outside the experiment. This approach provides insights into how memory interacts with stereotypes in real-world scenarios.

Second, unlike typical memory studies where subjects passively learn information followed by filler tasks, our design requires subjects to actively engage in memory encoding through a grading task. This task is time-consuming and involves active participation, making it a natural context for memory use. The accuracy in grading also allows us to verify if the information is correctly understood.

Third, our study examines both group- and individual-level beliefs and recalls. While stereotypes are inherently group-based, real-world experiences are often personal. By eliciting beliefs and recall for both levels, we assess how imperfect memory shapes the aggregation of personal experiences into group perceptions.

Our experiment yields the following results.

First, before the grading task, 92% of our subjects believe that male students are more knowledgeable about cars, which confirms the topic to be perceived stereotypically among our subjects.

Second, we find that relying on memory biases both posterior beliefs and recalls of the gender gap upwards only in the Female-advantaged class, where the stereotypes do not

accurately describe the true performance gap. This provides evidence that memory is not only imperfect, but also asymmetrically favors the information that is consistent with the stereotypes. If we look into the gender groups separately, we find a general inflated perception of the performance of both gender groups, while the males in the Female-advantaged class contribute to the larger deviation.

Third, we find no significant gender difference in beliefs and recalls for individual students, conditional on memory reliance. This implies that subjects do not simply forget stereotype-inconsistent student performances, i.e., female students with a high score and male students with a low score. However, when subjects need to rely on their memories, they tend to have more recall errors, leading to a moderate belief. Similarly, when we ask subjects to nominate the best student, there is no gender difference conditional on memory reliance.

Fourth, we design a nudge intervention, where subjects assess one individual student's performance before the group elicitation. This can be seen as increasing the salience and recall probability for that student, and hence may affect group beliefs. In the same experiment, we flip the order of group and individual elicitations for half of the subjects in the Hide condition. We find a limited effect of the nudge. We hypothesize that we lack sufficient statistical power to reject the null hypothesis, or the intervention is simply too gentle to induce a change.

In the experiment, we remain agnostic about the belief-updating process. To better understand the mechanisms driving our findings—particularly how imperfect memory influences the belief-updating process—we extend the standard Bayesian framework to incorporate imperfect recall. While Bayes' rule determines posterior beliefs based on prior beliefs and signals, our framework accounts for the stochastic nature of memory recall. Specifically, the decision-maker relies on recalled information rather than true information. However, recalled information is both noisy (recall uncertainty) and susceptible to contamination by default values (memory contamination), often assumed to be the prior beliefs in other contexts (Gabaix, 2019). These features reduce reliance on true information, a phenomenon we refer to as *memory attenuation*.

Applying this framework, we find strong evidence of memory attenuation: subjects rely 50% less on information from the graded quiz when they can only retrieve information from memory. We further decompose this effect into two key channels – memory contamination and recall uncertainty – and find evidence for both. Notably, memory contamination accounts for 72% of the observed attenuation. However, we do not find that subjects place greater weight on prior beliefs, suggesting they rely on defaults other than their prior beliefs.

Related literature. Our paper contributes to several strands of literature.

First, we contribute to the literature on stereotypes driven by inaccurate beliefs. Recent work highlights the role of asymmetric belief updating in reinforcing stereotypes (Reuben et al., 2014; Sarsons, 2022; Coffman et al., 2024; Eyting, 2024). Instead of varying the signal strength about individual performance, our study holds the information constant¹ and varies the reliance on memory. Furthermore, we do not

¹A recent paper by Coffman et al. (2025) shows asymmetry when information is endogenous.

impose assumptions about the belief updating rule in the experiment, as we only compare subjects receiving the same information. This allows us to isolate the effects of biased recall on stereotypical beliefs.

Second, our paper provides new empirical evidence of imperfect memory on belief formation. While recent theoretical models suggest that imperfect memory can lead to inaccurate beliefs (Bordalo et al., 2022), empirical evidence remains limited (for a review see Amelio and Zimmermann (2023)). Sial et al. (2024) argue that memory is biased based on correlational findings in the field, whereas we provide direct causal evidence to support this claim. Miserocchi (2023) provides the first empirical evidence of the link between memory and discrimination. She examines a setting where evaluators assess a single individual over time (“time series” information), while we focus on multiple anonymous individuals in a quiz (“cross-sectional” information). This distinction allows us to explore how memory operates differentially at the group and individual levels and affects decision-making. Graeber et al. (2024) examine the persuasiveness and decay of different forms of information, whereas our study focuses on how stereotypes can bias both recall and beliefs.

Third, we contribute to the growing literature on cognitive foundations in economics by linking imperfect memory with cognitive imprecision. These two mechanisms are often seen as distinct ways to simplify information processing (Enke, 2024). However, our study highlights their connection through the probabilistic nature of recall (Bordalo et al., 2022). We treat reliance on memory as an exogenous source of cognitive uncertainty while keeping the problem structure and available information fixed. Our findings reveal strong memory attenuation, consistent with broader behavioral attenuation results from Enke et al. (2024). We also introduce a novel way to manipulate cognitive uncertainty by adjusting memory reliance, which can be applied to study decisions in real-world scenarios.

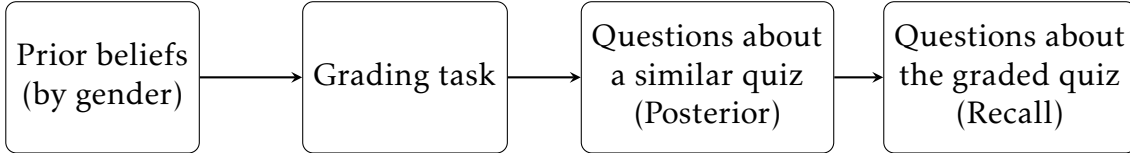
Fourth, we contribute to the literature on correlated beliefs, which examines the relationship between individual and group beliefs. Esponda et al. (2023) show that subjects place more weight on signals that are more representative of the associated group within a Bayesian framework. In contrast, our findings suggest that individuals rely on inaccurate recalls when forming beliefs, particularly for group beliefs. Additionally, Fan et al. (2023) demonstrate how simplifying heuristics can create inconsistencies between group and individual beliefs. Our findings show that group-level information is more difficult to recall, making group beliefs more susceptible to recall errors, whereas individual beliefs remain relatively more stable.

2 Experimental Design

We frame our study as a real effort grading task. Before and after the task, we elicit subjects’ beliefs about the performance of the students. We elicit their recalls to evaluate the use of memory at the end of the study. Both belief and recall elicitation are incentivized, with one randomly selected to determine the bonus of the subject. Figure 1 presents the timeline of the experiment. We use the grading task to encode memory for two reasons: (1) the grading process is natural and takes time; (2) the task creates homogeneous experiences for all subjects as the evaluation is objective and simple. In

the experiment, we vary between two classes with different gender performance gaps for subjects to grade, and the reliance on memory about the performance in the graded quiz after the grading. We also vary, in the same experiment, the order of elicitation to study the effect of nudging, which is detailed in Section 4. Full instructions are provided in the [Online Appendix](#).

Figure 1: Timeline of the experiment.



Now, we will first describe the student data and how the two classes are drawn.

2.1 Student data

We use the student data from the quiz in [Bordalo et al. \(2019\)](#). The quiz was conducted at a laboratory at the University of California, Santa Barbara (UCSB) in 2017. Students were mostly undergraduate students from UCSB, and we restricted our classes to those who completed high school in the US, which leaves us with 412 students. These students had to complete 20 multiple-choice questions for each of the six gender-stereotyped categories², without external help. All questions had to be answered and were incentivized. We inform our subjects of the background information of these students, and how the quiz was conducted.

We focus on the quiz about cars, as it is perceived to have the highest gender gap among the students. However, grading all 20 questions would be too time-consuming for our experiment. We therefore selected 10 questions that give the highest expected variance in performance conditional on gender. This ensures the greatest overlap in score distribution between males and females. Then, we randomly assign five questions for our subjects to grade and five questions for them to guess the performance, framed as a similar quiz. The first question in the graded quiz shown in Figure 3a serves as an example of these questions.

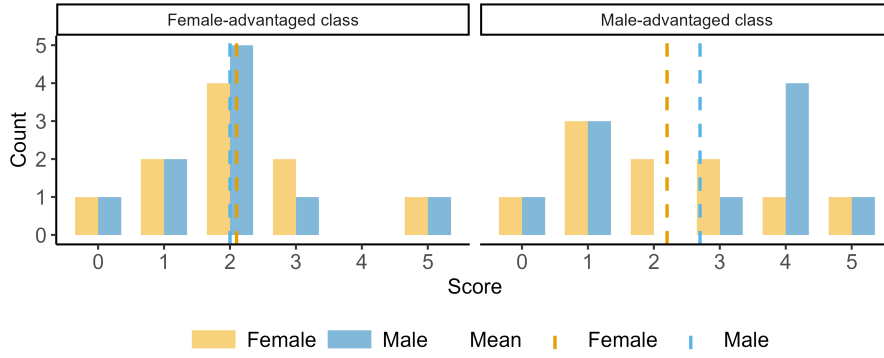
After fixing the questions, we drew 1,000 valid classes with the following two criteria: (1) there are exactly 10 males and 10 females, and (2) there are exactly one male and one female scored 0 and 5, respectively. These four students constitute the outliers in our experiment.

Appendix Figure 14 presents the sampling distribution. Among 1,000 valid classes meeting the two specified criteria, male students on average scored 0.21 more than female students, as indicated by the dotted line. For our experiment, we selected two classes representing the 20th percentile (Female-advantaged class) and the 80th percentile (Male-advantaged class) in terms of the gender gap. The gender gaps for the two classes are -0.1 and 0.5, respectively, as indicated by the solid lines in the figure.

Figure 2 illustrates the distribution and average scores by gender for each class. Each subject grades one of the two classes.

²Kardashians, Disney movies, cooking, cars, rock and roll, and video games.

Figure 2: Score distribution in the two classes.



Notes: The figure shows the score distribution in the Female- and Male-advantaged classes. The yellow bars represent females, and the blue bars represent males. The dotted lines represent the average score of the corresponding gender group.

2.2 The grading task

Before the grading task, subjects are informed of how the quiz was conducted. However, subjects do not know how many people took the quiz and how we selected the students for them to grade. When presented to the subject, each quiz script is associated with the gender of the student, and one of the randomly assigned top 10 most popular names for babies born during the 2000s with the corresponding gender ([Social Security Administration, 2024](#))³. We inform our subjects that, while the name is fictional, the gender information is accurate.

For each subject, we randomize the order of the quiz scripts, while we randomly place the four outliers at the 4th/8th/12th/16th positions. Additionally, we ensure the gender of the quiz scripts at the 4th and 16th position to be different, and elicit the beliefs and recalls for these two students after grading. The subject does not know these will be asked during the grading. We fix positions of outliers to maximize the power when we compare these individual measures.

The subject grades one submission at a time. Both at the top and bottom of the script, the subject sees the (fictional) name and (accurate) gender of the student. In the script, the answers from the student are displayed on the left, while the correct answers are displayed on the right (Figure 3a). The subject uses the checkbox on the right to indicate a correct answer. Once the subject finishes grading all five questions, a total score is automatically calculated.

After grading, the subject has to enter the score for the respective student in the corresponding cell on the student list at the bottom of the page (Figure 3b), before proceeding to the next student. The scores cannot be changed once the subject moves to the next student. We intentionally make the task manual, to strengthen (1) the association between the score and the student name; and (2) the salience of the gender group the student belongs to.

³The 20 names are displayed in Figure 3b.

Figure 3: Screenshots of the grading page

Grading Task (1/20)

Name: Ashley Gender: Female	Grading
The term "cabriolet" attached to a model name would indicate that _____. ☐ The car is an upscale luxury variant. ☒ The car is a convertible having a folding roof. ☐ The car is a sport-oriented performance variant. ☐ The car has only two doors. ☐ The car is European.	The car is a convertible having a folding roof. ☐
Which of the following boutique automakers originated in Sweden? ☐ Spyker ☐ Koenigsegg ☐ Pagani ☒ Wiesmann ☐ Bugati	Koenigsegg ☐

Name: Ashley **Gender:** Female **Total:** 1

Gender	Name	Graded	Name	Graded
Male	Andrew		Joseph	
Male	Christopher		Joshua	
Male	Daniel		Matthew	
Male	Ethan		Michael	
Male	Jacob		William	

Gender	Name	Graded	Name	Graded
Female	Ashley	1	Isabella	
Female	Elizabeth		Jenny	
Female	Emily		Madison	
Female	Emma		Olivia	
Female	Hannah		Samantha	

(a) Student submission

(b) Score entry

2.3 Reliance on memory

In addition to the class variation, we randomly assign subjects to one of the two displays for the grading records: Hide and Show. Under the Hide condition, subjects only see a checkmark (✓), and have to rely on their memory afterward. Under the Show condition, subjects can see the graded score for each student immediately after grading. Figure 4 shows an example for both conditions when the subject completed all 20 gradings. The grading record is displayed on the same page when we elicit beliefs and recalls after the grading task.

We assign two-thirds of the subjects to the Hide condition. For these subjects, we introduce an additional layer of randomization in the elicitation order. This randomization is designed to study the effects of a simple nudging intervention, which is explained further in Section 4.

Figure 4: Grading record displays in the Show and Hide conditions

(a) Show

Gender	Name	Graded	Name	Graded
Male	Andrew	1	Joseph	2
Male	Christopher	4	Joshua	1
Male	Daniel	0	Matthew	3
Male	Ethan	2	Michael	4
Male	Jacob	5	William	4

(b) Hide

Gender	Name	Graded	Name	Graded
Female	Ashley	1	Isabella	1
Female	Elizabeth	2	Jenny	5
Female	Emily	0	Madison	2
Female	Emma	2	Olivia	1
Female	Hannah	2	Samantha	2

2.4 Beliefs and recall elicitations

Throughout the study, subjects answer 12 incentivized questions, with one randomly selected to determine their bonus of up to £1. During the elicitations, subjects view one of two displays of the grading record, as shown in Figure 4, depending on their assigned condition.

Before the grading task, we elicit subjects' prior beliefs by asking them to estimate the average scores for males and females on separate pages. After the grading task, a similar quiz is introduced. The similar quiz also consists of 5 car-related multiple choice questions, and was also conducted by [Bordalo et al. \(2019\)](#) in the same format. We then ask subjects to make predictions for a similar quiz regarding (1) the average performance of the two gender groups, and (2) the performance of the two randomly selected outliers from different gender groups. Following these predictions, the subjects

nominate the student they believe performs the best. Finally, subjects are asked to recall the performance of the students and answer the same set of questions for the graded quiz. For all recall elicitation (except the nomination for the best performer), subjects have the option to check a box labeled “I don’t recall.” Selecting this option results in a zero payoff, aligning with the conventions in memory literature (Zimmermann, 2020). However, this option is rarely selected (10% among subjects in the Hide condition). Consequently, we exclude subjects who choose this option in our analysis of these recall outcomes.

Each of these questions is payoff-relevant, and followed by an unincentivized cognitive uncertainty measure similar to Enke and Graeber (2023). If subjects check ‘I don’t recall’, there is no need to answer the cognitive uncertainty question for that particular recall question.

Group-level measures. For the group-level measures, we randomize whether subjects are first asked about males or females, and this order remains consistent for each subject throughout the study. Subjects provide their answers using a slider ranging from 0 to 5, with intervals of 0.05. If this question is selected, subjects receive a bonus if the difference between their answer and the actual result is less than 0.2.

Individual-level measures. We elicit posterior beliefs and recalls only for the two outliers, positioned at the 4th and 16th places, from different gender groups. This focus on outliers is based on the assumption that they are more likely to be recalled, which helps reduce noise in the data. During elicitation, we randomize the order of the two outliers. To ensure comparability with the group-level measures, subjects provide their answers using a slider to select an integer between 0 and 5. If this question is selected, subjects receive the bonus only if their answer is correct.

Nominations. Subjects have two opportunities to nominate the best performer during the study. The first nomination occurs after the belief elicitation, where subjects select the best performer in a similar quiz. The second nomination takes place after the recall elicitation, where subjects choose the best performer in the graded quiz. If a nomination question is selected to determine the bonus, subjects earn £0.2 for each correct answer provided by the nominated student in the corresponding quiz, with a maximum potential bonus of £1 if the student answers all questions correctly.

2.5 Implementation and incentives

A total of 472 subjects from the US were recruited via Prolific after passing a simple attention check in December 2024. The experiment was programmed using oTree (Chen et al., 2016). The median completion time was approximately 25 minutes, and subjects earned an average of £3.30 from the study.

In addition to a participation fee of £2.56⁴, subjects were incentivized based on both grading accuracy and one of the 12 elicitation questions. To ensure accurate memory encoding, subjects earned £0.50 for correctly grading two randomly selected quiz

⁴The initial participation fee was set at £2.40. However, the actual time required exceeded both our expectations and the pilot study experience. As a result, we increased the participation fee after data collection to meet the platform’s requirements. Due to funding constraints, we also slightly reduced the sample size.

scripts. For the 12 elicitation questions, one was selected at random, and subjects earned an additional £1 if their answer was correct.

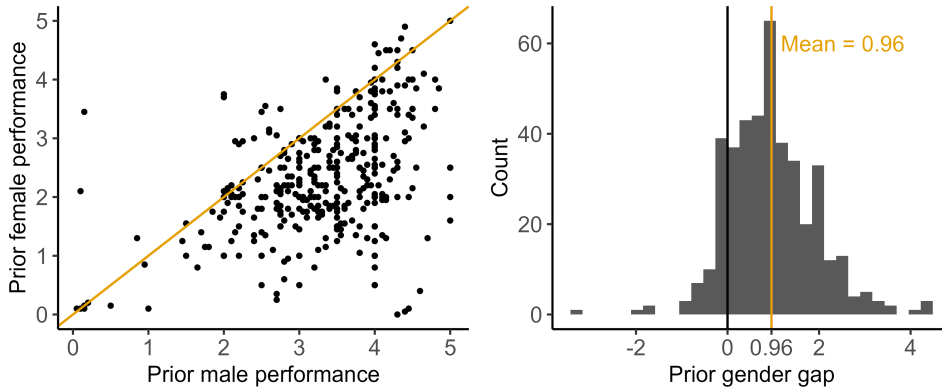
3 Experimental Results

For the analysis, we follow our pre-registration, and focus on the 420 subjects who (1) passed two attention checks, and (2) graded more than 15 students accurately.

3.1 Prior beliefs and stereotypes

We first confirm the presence of stereotypes among our subjects. Before the grading task, 92% of the subjects believe that male students achieve higher scores than female students on average. Figure 5 provides two visualizations of these prior beliefs. In the left panel, the scatterplot shows that most subjects fall below the orange 45-degree line, indicating a general perception that male students outperform female students. This supports the existing finding that males are perceived to excel in this domain.

Figure 5: Prior beliefs on the gender gap



Notes: The figure shows the prior beliefs about student performance across the gender groups (left panel) and about the gender gap (right panel). The gender gap is defined as the difference between the average male performance and the average female performance. In the left panel, the orange solid line represents the 45-degree line, and each dot represents one observation. In the right panel, the orange solid line represents the average prior beliefs about the gender gap, and the black solid line represents no gender gap.

To further illustrate the perceived gender gap, the right panel presents a histogram of these differences. On average, subjects estimate that male students answer 3.31 questions correctly, while female students answer 0.96 fewer, achieving an average of 2.34 questions. This implies that the probability of a male student answering a question correctly is 19% higher than that of a female student, which is comparable to the 15% reported in [Bordalo et al. \(2019\)](#). Notably, both the perceived average performance and the gender difference are higher than the actual scores derived from our 1,000 valid classes (Figure 14) and the original study of 412 quiz-takers. This pattern of overestimation is consistent with the findings of [Bordalo et al. \(2019\)](#).

One possible explanation for this overestimation of student performance is that subjects are shown the correct answers when introduced to the quiz but do not attempt the

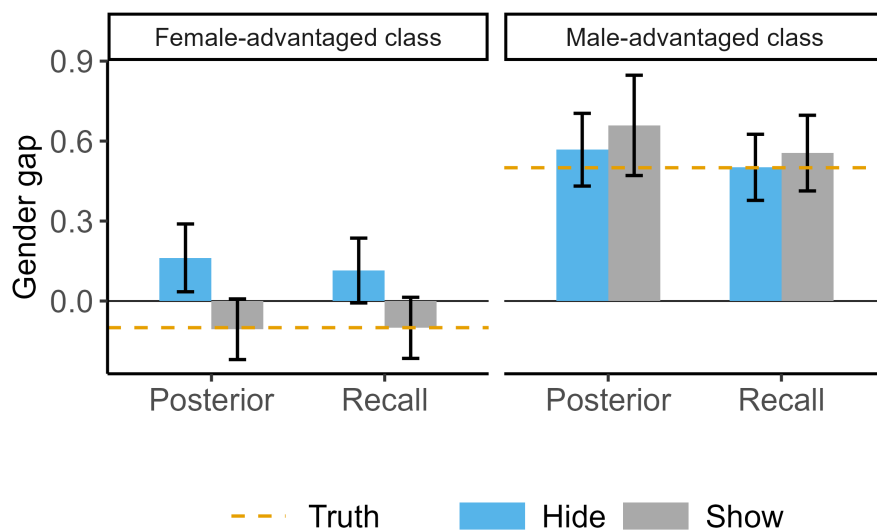
quiz themselves. Consequently, they may underestimate the quiz difficulty, leading to inflated perceptions of student performance.

3.2 The gender gap

We begin by illustrating how stereotypes influence the recall of past performance and posterior beliefs. Figure 6 shows the posterior beliefs and recall of the gender gap in the Hide (blue bars) and Show (gray bars) conditions across two classes. Across both conditions, subjects adjust their beliefs downward from their prior beliefs of 1.03 in response to the observed differences between their prior beliefs and the true gender gap. This downward adjustment suggests that the graded quiz provides informative feedback about students' performance.

When the grading record is shown, subjects adjust their beliefs proportionally, accurately calculating the difference in group averages and aligning their beliefs closely with the true gender gap. However, when the grading record is hidden, distinct patterns emerge between the Female- and Male-advantaged classes. For the Male-advantaged class, subjects report beliefs similar to those in the Show condition. In contrast, for the Female-advantaged class, subjects in the Hide condition predict the gender gap to be 0.26, which is higher than subjects in the Show condition (-0.01). Since subjects in both conditions grade the same class, this discrepancy is not driven by differences in information or stereotypes alone, but by the interaction between stereotypes and imperfect memory.

Figure 6: Perceived gender gap in beliefs and recall



Notes: The figure shows the perceived gender gap in the Hide (blue) and Show (gray) conditions, divided by the Female- and Male-advantaged classes. The gender gap is defined as the difference between the average male performance and the average female performance. The orange dotted line represents the true gender gap in the corresponding class. The band in each bar shows the 95% confidence interval.

This asymmetric pattern in beliefs is primarily driven by inaccuracies in recall. Specifically, subjects recall the gender gap as 0.114, rather than the true gender gap of -0.1 . Although the recalled gender gap is significantly different from the true gender gap

($p < 0.05$), but does not significantly differ from a zero gender gap ($p \simeq 0.178$). The perceived gender gap of 0.114 represents approximately 5% of the recalled average score for females. These results indicate that memory contamination by stereotypes influences recall and subsequent belief formation. Formal statistical evidence is provided in Table 2.

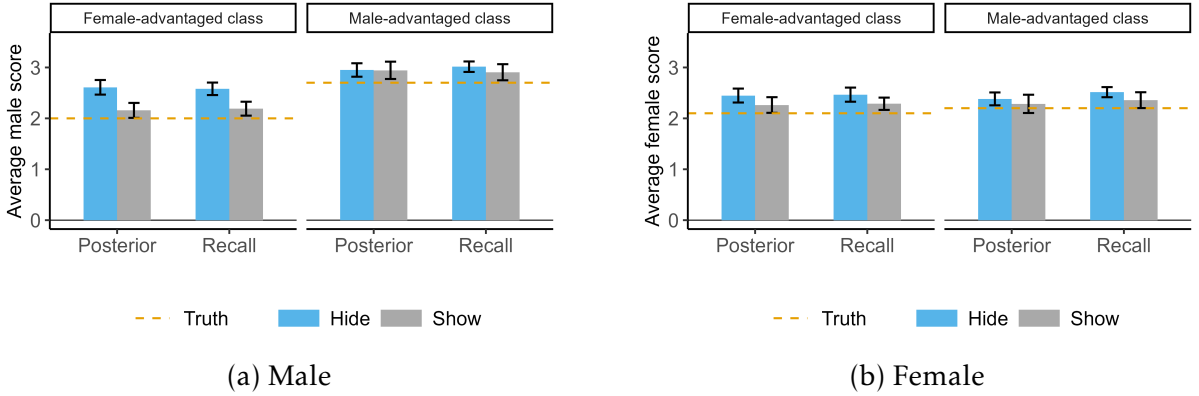
Qualitative evidence suggests that subjects from both the Show and Hide conditions use the results from the graded quiz to make predictions, while memory plays a significant role only in the Hide condition. Before the recall questions, we ask subjects to explain how they predict the performance for a similar quiz. After removing standard stop words and applying lemmatization, terms related to grading records, such as “previous” and “similar,” appear frequently in both conditions (Show: 16%; Hide: 18%). However, memory-related words like “remember,” “recall,” or “memory” are mentioned by 16.5% of subjects in the Hide condition, compared to just 0.7% in the Show condition. Figure 15 presents the word clouds by condition.

Result 1. *In the class that does not align with prior beliefs, subjects relying solely on memory update their posterior beliefs less, resulting in the persistence of stereotypes due to inaccurate recall.*

3.3 Inflated perceptions of group performance

As the gender gap represents the difference between the two gender groups, Result 1 can be driven by inaccurate beliefs and recalls in either group. Figure 7 presents the posteriors and recall patterns of the gender gap for each gender group.

Figure 7: Perceived group performance



Notes: The figure shows the posteriors and recall of the perceived average performance in the Hide (blue) and Show (gray) conditions for males in Panel (a) and females in Panel (b), divided by the Female- and Male-advantaged classes. The orange dotted line represents the true average score for the corresponding gender group and class. The band in each bar shows the 95% confidence interval.

For all subjects, recalled scores for both genders exceed the actual average quiz scores, indicating a failure to compute the true gender averages, even when detailed graded scores are provided. Notably, for subjects grading the Male-advantaged class or with detailed graded scores shown, this computational bias is not gender-specific, leading to an accurately recalled gender gap (see Figure 6).

In contrast, Result 1 is driven by a much stronger positive recall bias for male students

in the Female-advantaged class when the grading record is hidden. Specifically, the subjects recall the true average score of 2.0 as 2.58, which is 29% higher than the actual score, comparing to less than 17% for female students.

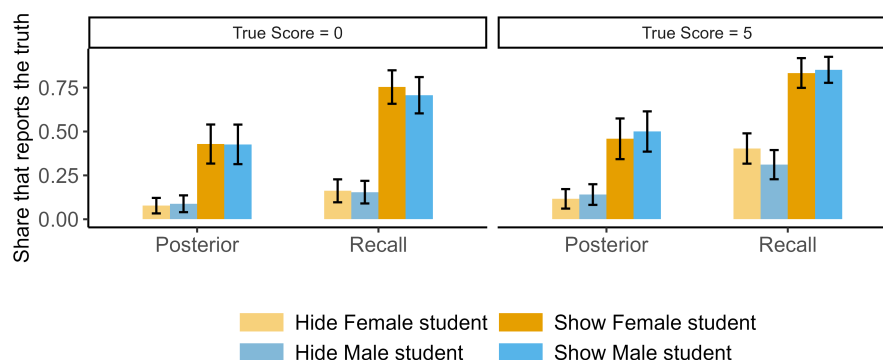
Formal statistical evidence is provided in Table 3.

Result 2. *Subjects tend to recall average gender performance more positively than the actual scores, regardless of gender. However, this overestimation is most pronounced for those with a Female-advantaged class and a hidden grading record.*

3.4 Perceptions of individual performance

Next, we examine the subjects' posteriors and recall for individual outliers, i.e., students with a score of 0 (worst score) or 5 (perfect score). Figure 8 illustrates the share of subjects who accurately reported scores of the male (blue bars) and female (orange bars) outliers in the Hide (light color) and Show (dark color) conditions. We categorize the outliers by their true scores.

Figure 8: Perceived performance of students with a score of 0 or 5



Notes: The figure shows the share of subjects who accurately reported scores of 0 (left panel) and 5 (right panel) for the male (blue bars) and female (orange bars) outliers in the Hide (light color) and Show (dark color) conditions. The band in each bar shows the 95% confidence interval.

When the subjects can only rely on memory, there is a higher recall error, with convergence towards the mean. The convergence further moderates the posteriors for these individuals. This suggests that the recall for individual performance is also imperfect and translates into beliefs. However, both the recall error and the difference between beliefs and the true scores are symmetric across genders.

Table 4 uses male students with a perfect score in the Show condition as the baseline. The estimated coefficient for Hide is the only one that remains significant across all specifications, indicating that subjects are less likely to recall information accurately (Columns 3–4) and exhibit larger recall errors (Columns 7–8). Columns 1–2 and 5–6 report similar variables based on posterior beliefs, for which no objective benchmark exists. However, the coefficients for both Female and the interaction term are close to zero, suggesting no gender difference once true performance is accounted for.

Result 3. *Subjects' beliefs and recalls for the outliers do not differ by gender.*

This result helps explain Result 1, which states that when subjects rely solely on

memory and encounter a class inconsistent with stereotypes, they update their beliefs less. One possible explanation is that subjects tend to forget student performances that contradict gender stereotypes. However, this appears inconsistent with Result 3, which suggests that recall accuracy and belief formation for individual students do not depend on gender or stereotype consistency.

Additionally, we consider how the reliance on memory may affect real decisions in this context: talent selection, i.e., nominating one single best performer among the students graded. As shown in Table 5, we find no gender difference in nominating the best performer on both Hide and Show conditions. Subjects in the Hide condition are less likely to identify or recall the best performer, but this effect is not gender biased. This suggests that the reliance on memory increases cognitive uncertainty, supporting our modeling of recalled information with cognitive noise in Section 5.

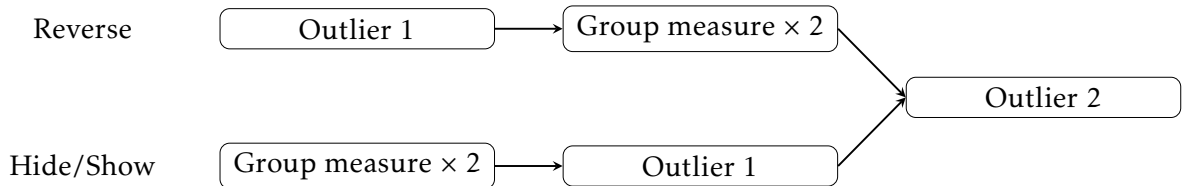
4 Nudging by Individual

As demonstrated in the previous section, the beliefs and recalls for individual outliers do not systematically vary by gender. Prompting subjects to first consider one of these outliers may influence their subsequent beliefs and recalls for the gender gap.

This phenomenon can also be interpreted within the memory literature as a memory rehearsal (Conlon, 2024). Memory rehearsal occurs when subjects retrieve information about an individual while answering a question related to an outlier. This process reinforces the memory, making it easier to recall later. Consequently, the rehearsal of outlier-related information may facilitate its integration when subjects retrieve memories to form beliefs in subsequent group-level measures.

In the same experiment, we randomize the order of elicitation for subjects with hidden grading records (as shown in Figure 9). For half of the subjects (*Reverse*), we elicit the perceived performance of one of the two outliers assigned, before eliciting the group-level measures.

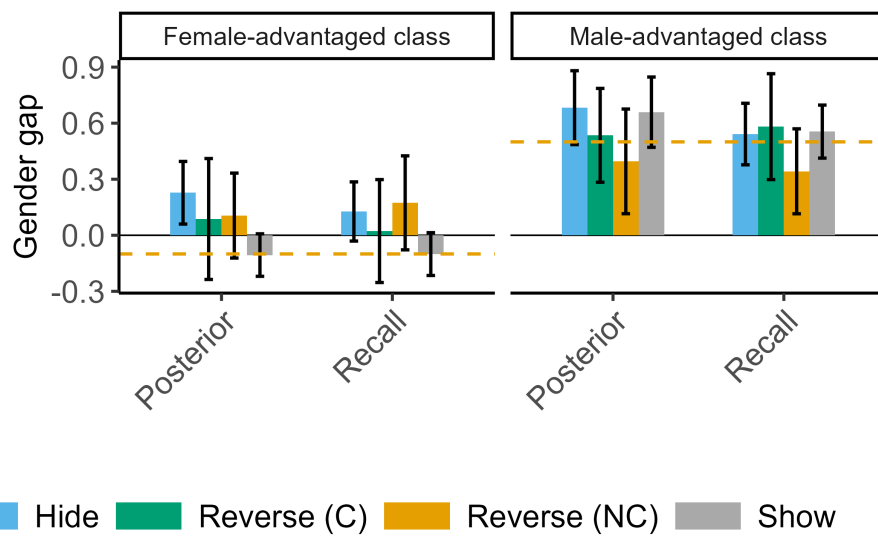
Figure 9: Elicitation order as a nudge.



Within the *Reverse* condition, subjects were further divided based on the stereotype alignment of the outlier. An outlier was classified as congruent (C) if it aligned with the stereotype, i.e., a male student achieving a perfect score or a female student receiving a zero score. Otherwise, an outlier was classified as non-congruent (NC). If the nudge is effective, subjects should demonstrate stronger stereotypical beliefs and recall when the nudge is congruent, and weaker stereotypical tendencies when the nudge is non-congruent.

Figure 10 and Table 6 demonstrate the effect of the nudge intervention. The results indicate that prompting subjects to an outlier does not change the perceived gender

Figure 10: Perceived gender gap after the nudge



Notes: The figure shows the posteriors and recall of the gender gap in the Hide (blue), Reverse (C) (green), Reverse (NC) (orange), and Show (grey) conditions, divided by the Female- (left) and Male-advantaged (right) classes. The gender gap is defined as the difference between the average male performance and the average female performance. In the Hide and Show conditions, group beliefs and recall are elicited first. In the Reverse (C) condition, beliefs and recall about either a male student with a perfect score or a female student with a zero score are elicited first. In the Reverse (NC) condition, beliefs and recall about either a male student with a zero score or a female student with a perfect score are elicited first. The orange dotted line represents the true gender gap in the corresponding class. The band in each bar shows the 95% confidence interval.

gap for both quizzes significantly.

Result 4. *Nudging subjects to first consider an outlier has minimal impact on altering their group-level beliefs.*

5 Structural Decomposition

In earlier sections, we remain agnostic about the belief updating process. Our experimental design isolates the effect of prior beliefs and their interaction with recall by exogenously manipulating reliance on memory and the information available (gender performance gap in class). Prior beliefs play a central role in standard economic models, where Bayes' rule determines posterior beliefs given prior beliefs and signals. But what happens when the decision maker (DM) must first recall these signals? In this section, we extend the Bayesian framework to incorporate imperfect recall and quantify the effects of prior beliefs.

This framework captures two key mechanisms when DM can only rely on imperfect recall: first, recall can be contaminated by some default values (memory contamination); second, recall is uncertain (recall uncertainty). Both mechanisms lead to a lower weight on the true score. If the default value for recall is the prior beliefs, both mechanisms shrink the posterior beliefs toward the prior beliefs (memory attenuation).

5.1 A simple Bayesian framework

DM grades a quiz for $2N$ students, consisting of N students from each gender group $g \in \{M, F\}$. The score for the quiz is a noisy measure of the ability a_{ig} of student i from gender group g , where the noise ϵ_{ig} is drawn from an independent normal distribution $\mathcal{N}(0, \sigma_\epsilon^2)$. The score is expressed as:

$$s_{ig} = a_{ig} + \epsilon_{ig} \quad (1)$$

DM is interested in knowing the student ability a_{ig} , which is drawn from an independent normal distribution $\mathcal{N}(a_g, \sigma_{a_g}^2)$, where a_g and $\sigma_{a_g}^2$ are the known gender-specific mean and variance, respectively.

The prior belief about the average score for gender group g is given by

$$\frac{1}{N} \sum_{i=1}^N s_{ig} = \bar{s}_g \sim \mathcal{N}\left(a_g, \frac{\sigma_g^2}{N}\right) \quad (2)$$

where $\sigma_g^2 := \sigma_{a_g}^2 + \sigma_\epsilon^2$. However, as we have shown earlier, recall is imperfect. Instead of using the actual scores s_{ig} for student i from gender group g , subjects rely on the recalled score r_{ig} , subject to additional cognitive noise $e_{ig} \sim \mathcal{N}(0, \sigma_e^2)$

$$r_{ig} = \mu_g s_{ig} + (1 - \mu_g) d_g + e_{ig} \sim \mathcal{N}(\mu_g s_{ig} + (1 - \mu_g) d_g, \mu_g^2 \sigma_g^2 + \sigma_e^2) \quad (3)$$

where d_g is the constant default score depending on gender g . The ability of the DM to recall is captured by $\mu_g \in [0, 1]$. The ability to recall is perceived as perfect by DM, i.e., $\tilde{\mu}_g = 1$, otherwise he can back out the true score from the recalled score.

We set the recalled score as a convex combination between the true score, and the default score d_g , which is usually assumed to be the mean of the prior belief a_g . When $\mu_g = 1$ and $\sigma_e^2 = 0$, recall is perfect. If recall is unbiased but imprecise, $\mu_g = 1$ and $\sigma_e^2 > 0$.

Consequently, the recalled average score for gender group g follows a normal distribution:

$$\frac{1}{N} \sum_{i=1}^N r_{ig} = \bar{r}_g \sim \mathcal{N}\left(\mu_g \bar{s}_g + (1 - \mu_g) d_g, \frac{\mu_g^2 \sigma_g^2 + \sigma_e^2}{N}\right) \quad (4)$$

Suppose the DM uses the mean prior beliefs as the default, i.e., $d_g = a_g$, the mean gender gap recalled is:

$$\tilde{\mathbb{E}}[\bar{r}_M - \bar{r}_F] = \mu_M \bar{s}_M - \mu_F \bar{s}_F + (1 - \mu_M) a_M - (1 - \mu_F) a_F \quad (5)$$

For DM with perfect memory, the posterior belief about the average ability $\bar{a}_g$ for gender group g is normally distributed, with the posterior mean given by:

$$\frac{\sigma_{a_g}^2}{\sigma_g^2} \cdot \bar{s}_g + \frac{\sigma_e^2}{\sigma_g^2} \cdot a_g \quad (6)$$

For DM with imperfect memory, with recall score vector $\mathbf{r}$ for all $2N$ students, the posterior belief about the average ability $\bar{a}_g$ for gender group g is normally distributed, with the posterior mean given by:

$$\lambda_g r_g + (1 - \lambda_g) a_g = \lambda_g \mu_g \bar{s}_g + \lambda_g (1 - \mu_g) a_g + (1 - \lambda_g) a_g = \lambda_g \mu_g \bar{s}_g + (1 - \lambda_g \mu_g) a_g \quad (7)$$

where $\lambda_g = \frac{\sigma_{a_g}^2}{\sigma_g^2 + \sigma_e^2}$. The posterior belief about the gender gap is then:

$$\tilde{\mathbb{E}}[\bar{a}_M - \bar{a}_F] = \lambda_M \mu_M \bar{s}_M - \lambda_F \mu_F \bar{s}_F + (1 - \lambda_M \mu_M) a_M - (1 - \lambda_F \mu_F) a_F \quad (8)$$

Together with the recalled gender gap in Equation 5, it is possible to estimate all four parameters ($\lambda_M, \mu_M, \lambda_F, \mu_F$) separately for the Hide and Show conditions, when there are sufficient independent variations in the average score for the two gender groups.

However, with the current experimental design, there are only two possible sets of true average scores by gender. Hence, we assume $\lambda\mu = \lambda_M \mu_M = \lambda_F \mu_F$, and simplify the expression for the posterior and recalled gender gap to:

$$\tilde{\mathbb{E}}[\bar{a}_M - \bar{a}_F] = \lambda\mu(\bar{s}_M - \bar{s}_F) + (1 - \lambda\mu)(a_M - a_F) \quad (9)$$

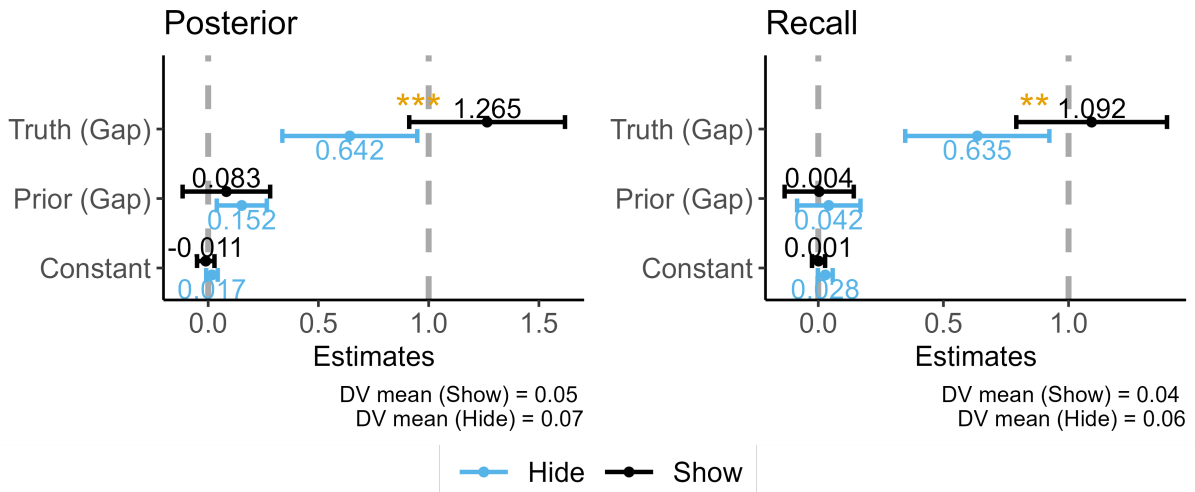
$$\tilde{\mathbb{E}}[\hat{s}_M - \hat{s}_F] = \mu(\bar{s}_M - \bar{s}_F) + (1 - \mu)(a_M - a_F) \quad (10)$$

The Hide condition in our experiment has two main effects. First, it reduces μ_g , as the recalled score becomes less correlated with the true score (memory contamination). Second, it reduces λ_g by increasing σ_e^2 , as subjects experience greater cognitive uncertainty regarding the recalled score (recall uncertainty). Both channels lead to a lower weight on the true score (memory attenuation).

5.2 Decomposition results

In this section, we normalize the full score to be 1, for all beliefs and recalls. We start by running the regression motivated by Equation 9 separately for Hide and Show conditions in Table 7. The estimates from the regression are reported on the left panel in Figure 11. The results indicate strong memory attenuation, i.e., subjects in the Hide condition are less responsive to the true gender gap. For 1% increase in the gender gap in the graded quiz, the posterior belief increases by 0.642%, instead of 1.265% in the Show condition. However, this reduction is not compensated by a statistically significant higher weight on the prior belief about the gender gap. One possible interpretation is that subjects' estimates converge toward zero, implying no perceived gender difference.

Figure 11: Gender gap decomposition



Notes: The figure shows the decomposition of the posterior beliefs (left) and recall (right) of the gender gap into the true gender gap in the graded quiz and the prior beliefs across the Hide (blue) and Show (black) conditions motivated by Equation 9 and 10. The band shows the 95% confidence interval. The stars indicate the statistical significance in comparing the coefficients in the Hide and Show conditions. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Next, we run a regression on the recalled gender gap, as motivated by Equation 10. The coefficient plot in the right panel of Figure 11 shows a similar pattern: subjects in the Hide condition are less sensitive to changes in the true gender gap, indicating memory contamination. A 1% increase in the true gender gap leads to only a 0.635% increase in the recalled gap, instead of 1.092% in the Show condition. We then examine the point estimates for prior beliefs. While recall appears slightly more sensitive to prior beliefs in the Hide condition, the effect is not statistically significant. This suggests that subjects may rely on other default scores when recalling information.

We then compute the estimates for both λ and μ and find that they are lower in the Hide condition. We interpret $\lambda\mu$ as the estimate for the true gender gap in the posterior regression, and μ as the estimate for the true gender gap in the recall regression. The estimated parameters are reported in the pooled column for the gender gap in Table 1. Alternatively, we can normalize the coefficients by the sum of all estimated coefficients in the corresponding regression, which yields qualitatively the same results, falling

between zero and one⁵. These findings confirm our hypothesis that both parameters are smaller when subjects can only rely on their memory. Memory contamination alone accounts for 72% of the memory attenuation effect.

Table 1: Estimated parameters

		Gender gap	Male performance	Individual performance
λ	Show	1.159	1.096	0.822
λ	Hide	1.011	0.706	0.581
μ	Show	1.092	1.03	0.826
μ	Hide	0.635	0.568	0.272
$\lambda\mu$	Show	1.265	1.129	0.679
$\lambda\mu$	Hide	0.642	0.401	0.158

Notes: The table shows the estimated parameters (μ, λ) across the Hide and Show conditions. The estimate for $\lambda\mu$ is obtained from the estimated coefficient for the true performance in the regression motivated by Equation 9. The estimate for μ is obtained from the estimated coefficient for the true performance in the regression motivated by Equation 10. The estimate for μ is the ratio between these two estimates.

We then conduct a similar regression for posterior beliefs and recall across gender groups. However, this analysis faces challenges due to limited variation in the true scores, so the results should be interpreted as suggestive. Figure 12 shows findings for male students, where greater variation exists, while results for female students, which are noisier, are presented in Figure 16. Formal estimation results for both gender groups are in Table 8.

As before, we observe memory attenuation when subjects rely solely on memory in the Hide condition. In the right panel, we find evidence of memory contamination by prior beliefs, as the estimated coefficients for prior beliefs are significantly positive in both conditions. However, the magnitudes do not differ across conditions, and subjects in the Hide condition do not rely more on prior beliefs when recalling the average male performance. Following the same parameter estimation method as for the gender gap, we find that both parameters are lower in the Hide condition (male performance column in Table 1).

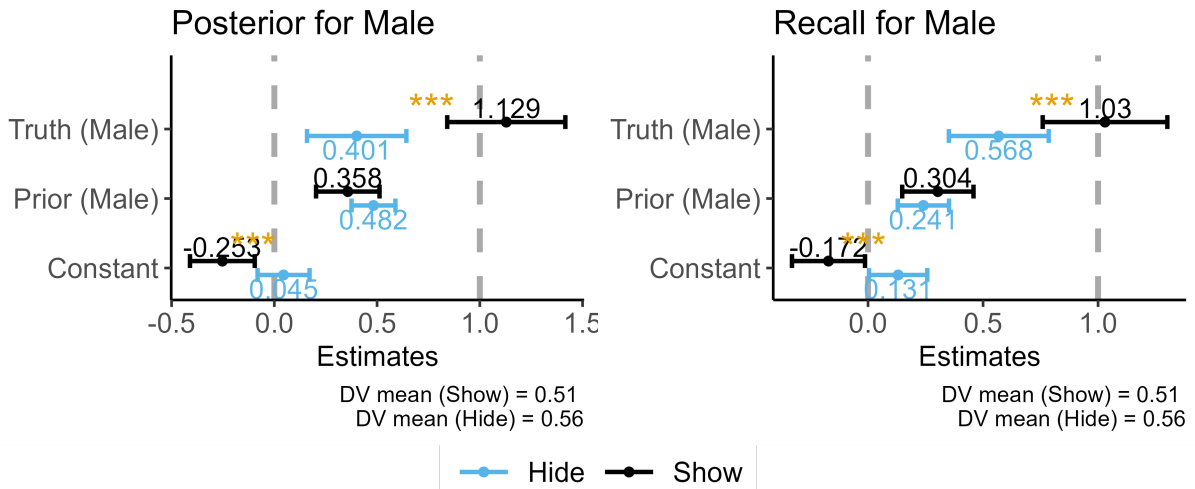
Finally, we examine posterior beliefs for individual students in Figure 13 and Table 9. We again observe memory attenuation, which is not offset by a higher weight on prior beliefs. However, the constant term in the Hide condition suggests that both posterior beliefs and recall converge toward a moderate score at around 2 out of 5, indicating attenuation in individual performance evaluation and recall. For the estimated parameters, we find that they are again lower for subjects in the Hide condition (individual performance column in Table 1).

5.3 Discussion

These findings are consistent with cognitive imprecision, but cognitive imprecision alone cannot fully explain the results. As shown in Figure 6, recall is also biased for the Female-advantaged class, suggesting that cognitive imprecision cannot account for this recall bias. Since most subjects overestimate the gender gap, our observations

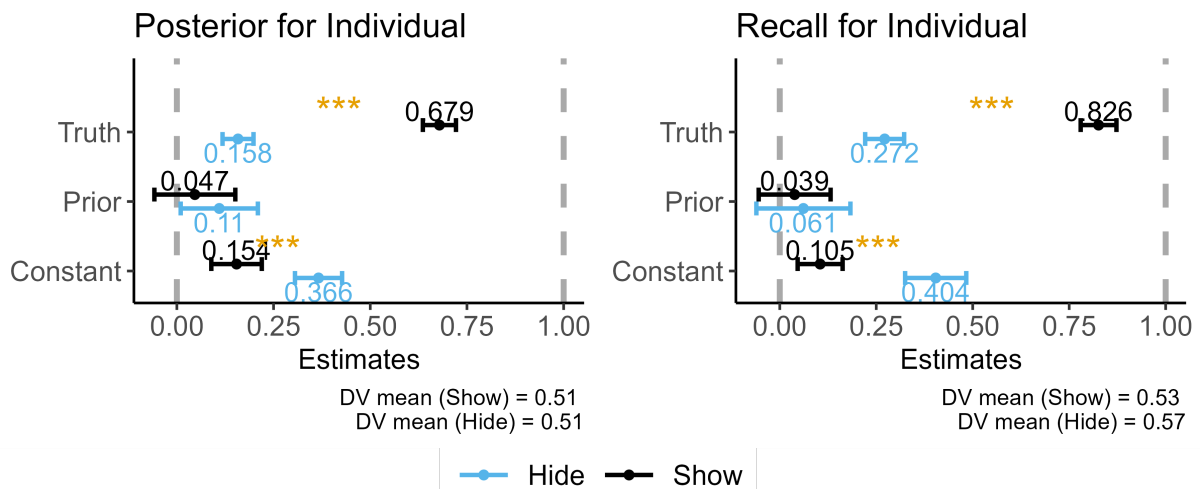
⁵The results are as follows: $\hat{\lambda}_{\text{Hide}} = 0.88$, $\hat{\lambda}_{\text{Show}} = 0.95$, $\hat{\mu}_{\text{Hide}} = 0.9$, $\hat{\mu}_{\text{Show}} = 0.995$.

Figure 12: Male performance decomposition



Notes: The figure shows the decomposition of the posterior beliefs (left) and recall (right) of the male performance into the true male performance in the graded quiz and the prior beliefs across the Hide (blue) and Show (black) conditions analogous to Equation 9 and 10. The band shows the 95% confidence interval. The stars indicate the statistical significance in comparing the coefficients in the Hide and Show conditions. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Figure 13: Individual performance decomposition



Notes: The figure shows the decomposition of the posterior beliefs (left) and recall (right) of the individual student performance into the true individual performance in the graded quiz and the prior beliefs across the Hide (blue) and Show (black) conditions analogous to Equation 9 and 10. The band shows the 95% confidence interval. The stars indicate the statistical significance in comparing the coefficients in the Hide and Show conditions. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

primarily capture belief updates in the downward direction. However, several aspects of the results suggest potential model misspecification.

First, the model predicts that the coefficients for the true score and prior belief should sum to one in both the Hide and Show conditions. This is not the case for subjects in the Show condition, where the two coefficients sum to 1.35 ($p < 0.1$ for equality with 1) for the gender gap and 1.38 ($p < 0.05$ for equality with 1) for average male performance.

Second, the current specification does not adequately capture recall in the Hide condition, leading to low R^2 . While it is intuitive that subjects might revert to a moderate score for individual recalls, this pattern is less clear for the gender gap and average male performance.

Third, substantial differences in constant terms for male performance suggest potential omitted variable bias. One possible explanation is memory interference, as discussed in the memory literature (Bordalo et al., 2022). We assume independent belief formation between gender groups, which restricts the coefficients to lie between zero and one, with their sum equal to 1. However, this can be relaxed if we allow for interference.

6 Conclusion

Each day, hiring managers review numerous CVs and conduct multiple interviews. While some managers take notes to document interactions, others rely on lasting impressions that shape their beliefs about individual candidates, influence their perceptions of future candidates with similar backgrounds, and ultimately impact hiring decisions.

Our experiment replicates this hiring process through a novel grading task centered on a gender-stereotypical quiz. We investigate how imperfect memory leads to sticky stereotypes.

We find that the reliance on memory distorts group beliefs and recall, particularly when the group does not align with the stereotype. The distortion is driven by excessive overestimation for male students, when subjects rely solely on memory and the stereotypes do not accurately describe the class performance.

However, this distortion does not influence individual evaluations when subjects possess some knowledge about the individual. This suggests subjects do not forget the individuals with performance inconsistent with the stereotype. Instead, subjects may perceive these correlated beliefs as different decisions. We then randomly emphasize individuals with extreme performance to nudge group beliefs and find limited effects.

Finally, by assuming the DM uses recalled information, instead of the true information, we distinguish two key channels of how imperfect memory affects belief updating: (1) recalled information being contaminated, most possibly by prior beliefs, and (2) lower certainty about recalled information. We look at three sets of metrics: gender gap, average score by gender, and randomly selected individual students with extreme performance. For all three sets of metrics, we find strong evidence for both mechanisms.

These findings shed light on why stereotypes persist and how they contribute to the continued prevalence of inaccurate beliefs, even when information is widely accessible. Efforts to reduce discrimination through belief correction should consider not only the immediate change in beliefs but also individuals' ability to recall the information. We also show that stereotypes can quickly fade when people have specific information about an individual, highlighting the importance of initial contact with disadvantaged groups in challenging stereotypes. However, these initial contacts are less effective in changing well-established stereotypes about the group as a whole.

References

- Aigner, Dennis J. and Glen G. Cain (1977) “Statistical Theories of Discrimination in Labor Markets,” *Industrial and Labor Relations Review*, 30 (2), 175.
- Amelio, Andrea and Florian Zimmermann (2023) “Motivated Memory in Economics—A Review,” *Games*, 14 (1), 15.
- Becker, Gary S. (1957) *The Economics of Discrimination*, Economic Research Studies, Chicago, IL: University of Chicago Press, <https://press.uchicago.edu/ucp/books/book/chicago/E/bo22415931.html>.
- Bohren, J. Aislinn, Kareem Haggag, Alex Imas, and Devin G. Pope (2023) “Inaccurate Statistical Discrimination: An Identification Problem,” *Review of Economics and Statistics*, 1–45.
- Bordalo, Pedro, Katherine Coffman, Nicola Gennaioli, Frederik Schwerter, and Andrei Shleifer (2021) “Memory and Representativeness,” *Psychological Review*, 128 (1), 71–85.
- Bordalo, Pedro, Katherine Coffman, Nicola Gennaioli, and Andrei Shleifer (2016) “Stereotypes,” *The Quarterly Journal of Economics*, 131 (4), 1753–1794.
- (2019) “Beliefs about Gender,” *American Economic Review*, 109 (3), 739–773.
- Bordalo, Pedro, John J Conlon, Nicola Gennaioli, Spencer Y Kwon, and Andrei Shleifer (2022) “Memory and Probability,” *The Quarterly Journal of Economics*, 138 (1), 265–311.
- Chen, Daniel L., Martin Schonger, and Chris Wickens (2016) “oTree—An open-source platform for laboratory, online, and field experiments,” *Journal of Behavioral and Experimental Finance*, 9, 88–97.
- Coffman, Katherine B., Scott Kostyshak, and Perihan O. Saygin (2025) “Choosing and Using Information in Evaluation Decisions,” Version Number: 1.
- Coffman, Katherine, Manuela R Collis, and Leena Kulkarni (2024) “Stereotypes and Belief Updating,” *Journal of the European Economic Association*, 22 (3), 1011–1054.
- Conlon, John (2024) “Memory Rehearsal and Belief Biases,” August, https://johnjconlon17.github.io/website/conlon_rehearsal_beliefs.pdf.
- Enke, Benjamin (2024) “The Cognitive Turn in Behavioral Economics,” September, https://benjamin-enke.com/pdf/Cognitive_turn.pdf.
- Enke, Benjamin and Thomas Graeber (2023) “Cognitive Uncertainty,” *The Quarterly Journal of Economics*, 138 (4), 2021–2067.
- Enke, Benjamin, Thomas Graeber, Ryan Oprea, and Jeffrey Yang (2024) “Behavioral Attenuation,” September.
- Esponda, Ignacio, Ryan Oprea, and Sevgi Yuksel (2023) “Seeing What is Representative,” *The Quarterly Journal of Economics*, 138 (4), 2607–2657.

- Eyting, Markus (2024) “Why Do We Discriminate? The Role of Motivated Reasoning,” https://meyting.com/documents/motivated_discrimination.pdf.
- Fan, Tony Q., Yucheng Liang, and Cameron Peng (2023) “The Inference-Forecast Gap in Belief Updating,” September, https://www.dropbox.com/sc1/fi/phv1pyovxnfqx8uks7n19/IFG_manuscript.pdf?rlkey=mbdjfs9di3x3fbxhlrvvv5cop&e=1&dl=0.
- Gabaix, Xavier (2019) “Behavioral inattention,” in *Handbook of Behavioral Economics: Applications and Foundations 1*, Vol. 2, 261–343: Elsevier.
- Graeber, Thomas, Christopher Roth, and Florian Zimmermann (2024) “Stories, Statistics, and Memory,” *The Quarterly Journal of Economics*, qjae020.
- Misrocchi, Francesca (2023) “Discrimination through Biased Memory,” November, https://www.dropbox.com/sc1/fi/80kt6gz8bke5lwc2g9uh/DiscriminationRecall_FM_2023.pdf?rlkey=gvp1n9o24m5nyrjo5u8yivpz&e=2&dl=0.
- Phelps, Edmund S. (1972) “The Statistical Theory of Racism and Sexism,” *The American Economic Review*, 62 (4), 659–661, <https://www.jstor.org/stable/1806107>.
- Reuben, Ernesto, Paola Sapienza, and Luigi Zingales (2014) “How stereotypes impair women’s careers in science,” *Proceedings of the National Academy of Sciences*, 111 (12), 4403–4408, Publisher: Proceedings of the National Academy of Sciences.
- Sarsons, Heather (2022) “Interpreting Signals in the Labor Market: Evidence from Medical Referrals,” September, <https://drive.google.com/file/d/12LI5b4Xg7D1NWt-m12qw-PaMH1ihd10V/view>.
- Sial, Afras, Justin R. Sydnor, and Dmitry Taubinsky (2024) “Biased Memory and Perceptions of Self-Control,” July, <https://www.dropbox.com/sc1/fi/okt5rbibltiptgpi64qm6/Limited-memory-and-self-control.pdf?rlkey=qw4b0tdtxaqjqqa6su2gp6vto&e=1&dl=0>.
- Social Security Administration (2024) “Top names of the 2000s,” March, <https://www.ssa.gov/oact/babynames/decades/names2000s.html>.
- Zimmermann, Florian (2020) “The Dynamics of Motivated Beliefs,” *American Economic Review*, 110 (2), 337–361.

A Regression Tables

Table 2: Deviation from gender gap in graded quiz

	<i>Dependent variable:</i>	
	Posterior	Recall
	(1)	(2)
Hide	0.27*** (0.09)	0.22** (0.09)
Male-advantaged Class	0.16 (0.11)	0.06 (0.09)
Hide $\times$ Male-advantaged Class	-0.36** (0.15)	-0.27** (0.13)
Constant	-0.01 (0.06)	-0.001 (0.06)
Observations	420	401

Notes: Deviation is defined as the prediction/recall minus the graded quiz statistics. Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table 3: Deviation from gender gap in graded quiz

	<i>Dependent variable:</i>			
	Posterior	Recall	Posterior	Recall
	(1)	(2)	(3)	(4)
Hide	0.45*** (0.10)	0.39*** (0.09)	0.19* (0.11)	0.18* (0.09)
Male-advantaged Class	0.09 (0.11)	0.02 (0.11)	-0.08 (0.12)	-0.03 (0.10)
Hide × Male-advantaged Class	-0.45*** (0.15)	-0.28** (0.13)	-0.09 (0.15)	-0.02 (0.13)
Constant	0.16** (0.07)	0.19*** (0.07)	0.16** (0.08)	0.19*** (0.06)
Gender	Male	Male	Female	Female
Observations	420	404	420	404

Notes: Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table 4: Individual beliefs and recalls

	Dependent variable:							
	Same as record				Absolute deviation			
	Posterior (1)	Posterior (2)	Recall (3)	Recall (4)	Posterior (5)	Posterior (6)	Recall (7)	Recall (8)
Female	-0.07 (0.15)	-0.02 (0.10)	-0.05 (0.19)	-0.04 (0.10)	0.04 (0.17)	0.04 (0.07)	-0.01 (0.17)	0.13 (0.09)
Score == 0	-0.11 (0.26)	-0.03 (0.12)	-0.25 (0.21)	-0.11 (0.12)	0.46 (0.30)	0.08 (0.10)	0.64*** (0.24)	0.51*** (0.09)
Hide	-0.43** (0.19)	-0.26*** (0.09)	-0.64*** (0.21)	-0.45*** (0.10)	1.33*** (0.20)	1.16*** (0.07)	1.22*** (0.21)	1.29*** (0.09)
Female × Score == 0	-0.01 (0.30)	0.09 (0.15)	0.17 (0.27)	0.05 (0.14)	-0.24 (0.35)	-0.11 (0.14)	-0.59* (0.34)	-0.20 (0.13)
Hide × Female	0.04 (0.25)	-0.04 (0.11)	0.09 (0.28)	0.11 (0.12)	0.36 (0.28)	0.11 (0.10)	0.21 (0.31)	-0.36*** (0.12)
Hide × Score == 0	0.09 (0.33)	-0.07 (0.13)	0.09 (0.30)	-0.11 (0.13)	-0.19 (0.41)	0.12 (0.12)	0.30 (0.36)	0.21* (0.12)
Hide × Female × Score == 0	0.03 (0.41)	-0.04 (0.17)	-0.19 (0.38)	-0.12 (0.16)	-0.15 (0.49)	-0.16 (0.18)	0.19 (0.48)	0.28 (0.17)
Constant	0.55*** (0.13)	0.49*** (0.08)	0.92*** (0.15)	0.91*** (0.09)	0.54*** (0.14)	0.62*** (0.05)	0.17 (0.12)	0.04 (0.07)
Class	Male-advantaged	Female-advantaged	Male-advantaged	Female-advantaged	Male-advantaged	Female-advantaged	Male-advantaged	Female-advantaged
N. subject	209	211	205	206	209	211	205	206
Observations	418	422	396	405	418	422	396	405

Notes: All columns control for round fixed effects. Absolute deviation is defined as the distance between the prediction/recall from the record. Robust standard errors are in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table 5: Nominations for best performer

	<i>Dependent variable:</i>					
	Posterior			Recall		
	(1)	(2)	(3)	(4)	(5)	(6)
Hide	−0.26*** (0.05)	−0.26*** (0.05)	−0.26*** (0.05)	−0.27*** (0.05)	−0.27*** (0.05)	−0.27*** (0.05)
Female	−0.05 (0.08)	−0.05 (0.08)	−0.05 (0.08)	−0.11 (0.08)	−0.11 (0.08)	−0.11 (0.08)
Hide:Female	0.01 (0.09)	0.01 (0.09)	0.01 (0.09)	0.09 (0.09)	0.09 (0.09)	0.09 (0.09)
Hide:Score<5	0.29*** (0.05)	0.29*** (0.05)	0.29*** (0.05)	0.30*** (0.05)	0.30*** (0.05)	0.29*** (0.05)
Female:Score<5	0.04 (0.08)	0.04 (0.08)	0.04 (0.08)	0.11 (0.08)	0.10 (0.08)	0.10 (0.08)
Hide:Female:Score<5	−0.02 (0.09)	−0.02 (0.09)	−0.02 (0.09)	−0.09 (0.09)	−0.09 (0.09)	−0.09 (0.09)
Constant	0.49*** (0.04)	0.47*** (0.04)	0.48*** (0.04)	0.52*** (0.04)	0.49*** (0.04)	0.51*** (0.04)
Class FE	Yes	Yes	Yes	Yes	Yes	Yes
Student list order FE	No	Yes	Yes	No	Yes	Yes
Individual elicitation FE	No	Yes	No	No	Yes	No
Round FE	No	No	Yes	No	No	Yes
N. subject	420	420	420	420	420	420
Observations	8,400	8,400	8,400	8,400	8,400	8,400

Notes: Unit of observation is at the subject-student level. Robust standard errors are in parentheses.
 *p<0.1; **p<0.05; ***p<0.01.

Table 6: Deviation from gender gap in graded quiz after nudge (Hide only)

	<i>Dependent variable:</i>			
	Posterior		Recall	
	Female-advantaged	Male-advantaged	Female-advantaged	Male-advantaged
	(1)	(2)	(3)	(4)
Reverse (C)	−0.14 (0.19)	−0.15 (0.16)	−0.10 (0.16)	0.04 (0.17)
Reverse (NC)	−0.12 (0.14)	−0.29 (0.17)	0.05 (0.15)	−0.20 (0.14)
Constant	0.33*** (0.09)	0.18* (0.10)	0.23*** (0.08)	0.04 (0.08)
Observations	135	136	129	129

Notes: Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table 7: Structural estimation (Gap)

	<i>Dependent variable:</i>		
	Posterior	Recall	
	(1)	(2)	(3)
Recall (Gap)	0.89*** (0.09)		
Truth (Gap)		1.27*** (0.18)	1.09*** (0.15)
Prior (Gap)	0.08 (0.09)	0.08 (0.10)	0.004 (0.07)
Hide $\times$ Recall (Gap)	-0.15 (0.11)		
Hide $\times$ Truth (Gap)		-0.62*** (0.24)	-0.46** (0.21)
Hide $\times$ Prior (Gap)	0.05 (0.10)	0.07 (0.12)	0.04 (0.10)
Hide	0.003 (0.02)	0.03 (0.02)	0.03 (0.02)
Constant	-0.001 (0.01)	-0.01 (0.02)	0.001 (0.01)
Observations	401	420	401
R ²	0.51	0.15	0.13

Notes: Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table 8: Structural estimation (Level)

	<i>Dependent variable:</i>							
	Posterior				Recall			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Recall (Same)			0.67*** (0.11)	0.67*** (0.23)				
Truth (Same)	1.12*** (0.16)	0.22 (1.21)			1.13*** (0.15)	0.36 (1.18)	0.72 (1.00)	1.03*** (0.14)
Prior (Same)			0.16** (0.07)	0.17* (0.10)	0.36*** (0.08)	0.18* (0.10)	0.02 (0.09)	0.30*** (0.08)
Hide × Recall (Same)			−0.01 (0.13)	−0.01 (0.24)				
Hide × Truth (Same)	−0.64*** (0.22)	−0.88 (1.54)			−0.73*** (0.19)	−1.00 (1.49)	−0.22 (1.31)	−0.46*** (0.18)
Hide × Prior (Same)			0.15* (0.09)	−0.08 (0.11)	0.12 (0.10)	0.03 (0.11)	0.15 (0.11)	−0.06 (0.10)
Hide	0.35*** (0.10)	0.41 (0.66)	−0.08 (0.06)	0.05 (0.08)	0.30*** (0.10)	0.44 (0.64)	0.05 (0.56)	0.30*** (0.10)
Constant	−0.02 (0.08)	0.36 (0.52)	0.06 (0.05)	0.07 (0.07)	−0.25*** (0.08)	0.21 (0.51)	0.15 (0.42)	−0.17** (0.08)
Gender Group	Male	Female	Male	Female	Male	Female	Male	Female
Observations	420	420	404	404	420	420	404	404
R ²	0.13	0.01	0.52	0.38	0.34	0.07	0.05	0.28

Notes: Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

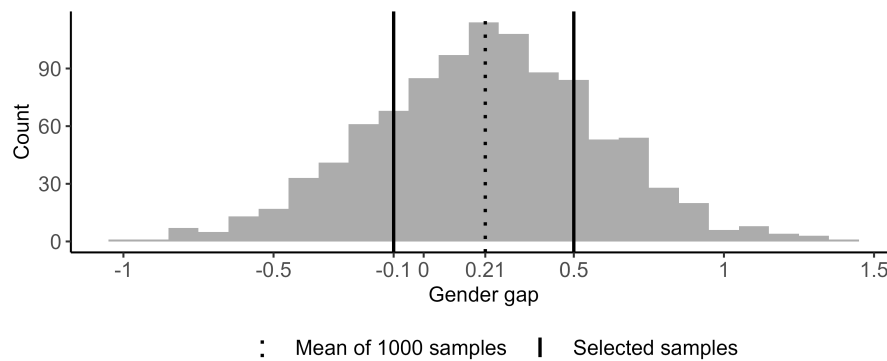
Table 9: Structural estimation (Individual)

	<i>Dependent variable:</i>							
	Posterior							Recall
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Recall		0.71*** (0.03)						
Truth	0.68*** (0.02)		0.68*** (0.02)	0.67*** (0.03)	0.69*** (0.03)	0.67*** (0.04)	0.65*** (0.04)	0.83*** (0.03)
Prior		0.01 (0.05)	0.05 (0.06)	0.07 (0.11)	0.01 (0.08)	0.14* (0.09)	−0.05 (0.09)	0.04 (0.05)
Hide × Recall		−0.21*** (0.04)						
Hide × Truth	−0.52*** (0.03)		−0.52*** (0.03)	−0.49*** (0.04)	−0.55*** (0.04)	−0.50*** (0.05)	−0.50*** (0.05)	−0.55*** (0.04)
Hide × Prior		0.06 (0.07)	0.06 (0.08)	0.04 (0.14)	0.09 (0.11)	−0.05 (0.12)	0.17 (0.13)	0.02 (0.08)
Hide	0.25*** (0.02)	0.05 (0.04)	0.21*** (0.05)	0.21** (0.10)	0.21*** (0.06)	0.28*** (0.06)	0.12 (0.09)	0.30*** (0.05)
Constant	0.18*** (0.02)	0.13*** (0.03)	0.15*** (0.03)	0.15* (0.08)	0.16*** (0.04)	0.10** (0.04)	0.22*** (0.07)	0.10*** (0.03)
Sample	Pool	Pool	Pool	Male	Female	C	NC	Pool
Observations	840	801	840	420	420	428	412	801
R ²	0.47	0.58	0.48	0.47	0.49	0.52	0.45	0.52

Notes: All measures are standardized. Standard errors are clustered at the subject level. C indicates that a male student with a score of 1 or a female student with a score of 0 is elicited first. NC indicates that a male student with a score of 0 or a female student with a score of 1 is elicited first. Robust standard errors are in parentheses. *p<0.1; **p<0.05; ***p<0.01.

B Additional Figures

Figure 14: The distribution of the gender gap



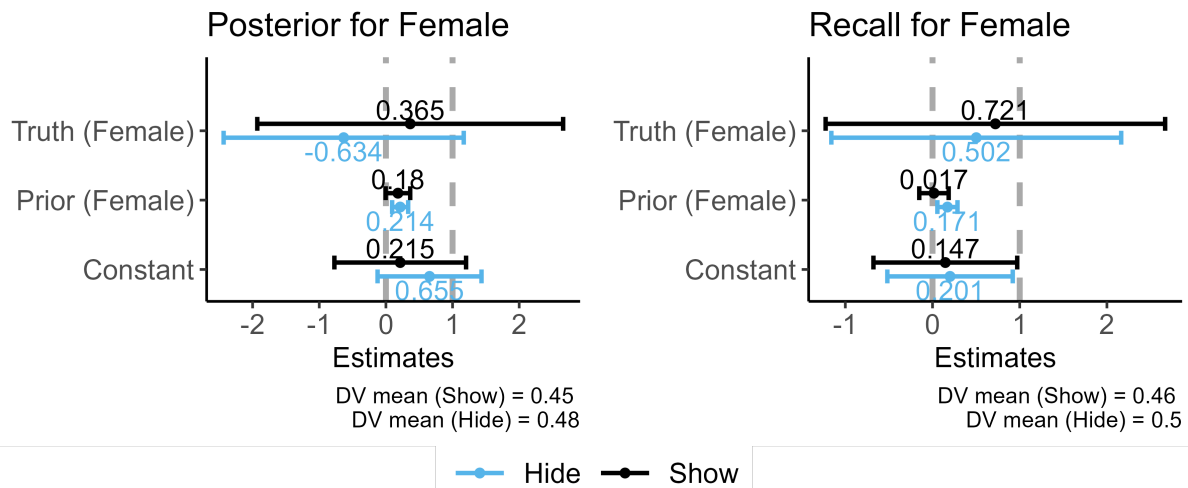
Notes: The figure shows the distribution of the gender gap in the 1,000 classes drawn from the student data in [Bordalo et al. \(2019\)](#). The gender gap is defined as the difference between the average male performance and the average female performance. In each of these classes, there are exactly 10 males and 10 females, and exactly one male and one female scored 0 and 5, respectively. The solid lines indicate the gender gap in classes in the 20th percentile (Female-advantaged class) and in the 80th percentile (Male-advantaged class). The dotted line indicates the mean in the gender gap among 1,000 valid classes.

The figure consists of two word clouds, (a) and (b), arranged horizontally. Both word clouds are shaped like a car. Word cloud (a) is titled '(a) Show' and contains many words, with 'score' and 'quiz' being the largest. Word cloud (b) is titled '(b) Hide' and contains fewer words, with 'score' and 'quiz' still being the largest, but 'remember' is highlighted in red.

(a) Show

(b) Hide

Figure 16: Female performance decomposition



Notes: The figure shows the decomposition of the posterior beliefs (left) and recall (right) of the female performance into the true female performance in the graded quiz and the prior beliefs across the Hide (blue) and Show (black) conditions analogous to Equation 9 and 10. The band shows the 95% confidence interval. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$